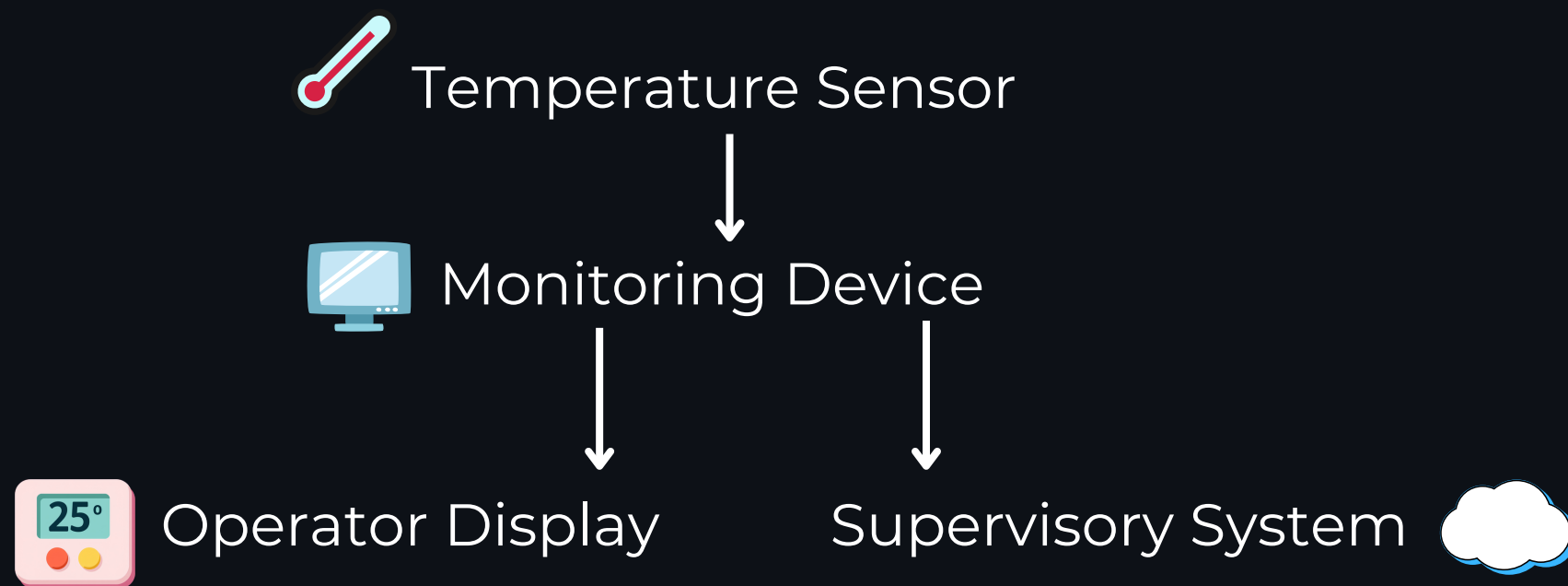

Reliability Pattern #1

Hardware Abstraction Layer (HAL)

Using a Temperature Monitoring Device as an example



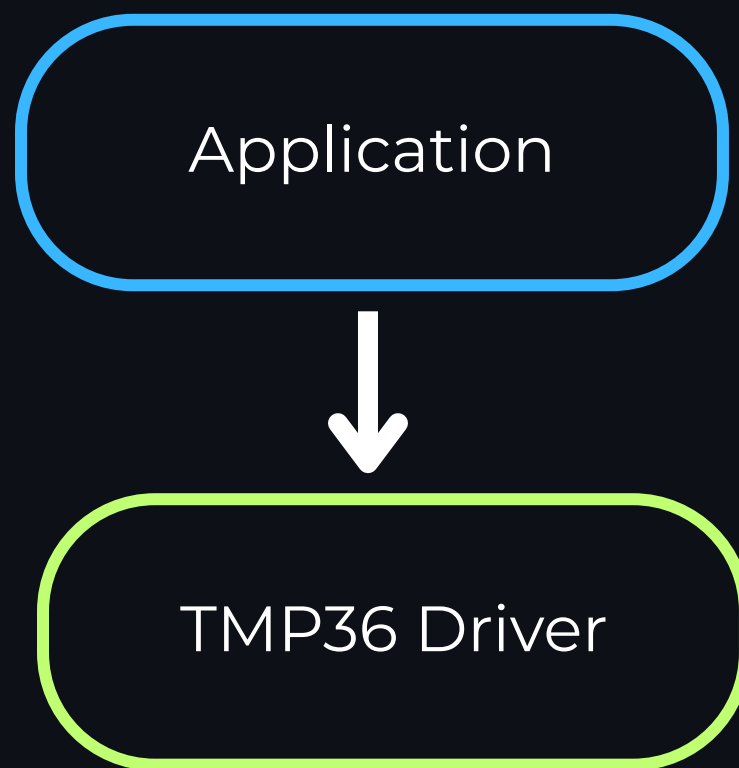
The System



Responsibilities:

- ▶ Read sensor
- ▶ Display temperature
- ▶ Report measurement

Common Design



Under it add:

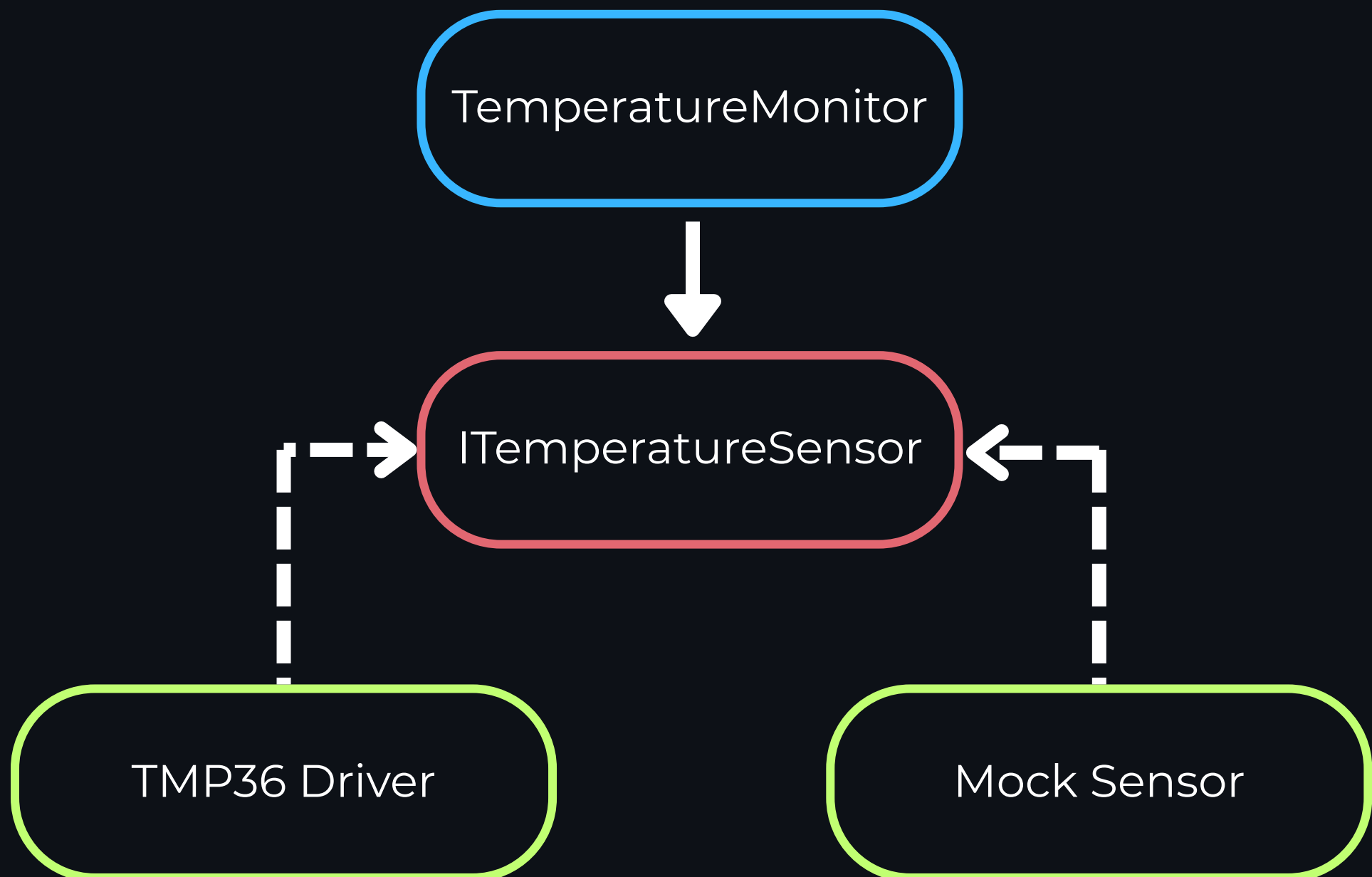
- ✗ Direct dependency on hardware
 - ✗ Difficult to test
 - ✗ Hardware change impacts application
-

What happen later?

- ▶ Sensor replacement ?
- ▶ New MCU ?
- ▶ Hardware failure simulation ?
- ▶ Unit testing ?
- ▶ Unexpected system state

All become difficult.

Introduce HAL



Benefits

- ✓ Hardware independence
- ✓ Better fault isolation
- ✓ Easier Testing
- ✓ Easier maintenance
- ✓ Reusable software

Want the Full Design?

GitHub contains:

- ✓ UML diagram
- ✓ C++ source code
- ✓ Design trade-offs

Repository link in comments

Coming Next....

Part 3:

Retry Mechanism:

- ▶ Increase resilience by automatically recovering from transient failures.
- ▶ Discover design considerations and real-world implementation